Every step shown | File 7 / 7

ANS Chapter 1 full worked answers

الإجابات الكاملة: افتحه بعد حل ملف الأسئلة

Each answer gives the result, the reason a marker looks for, and in the `why` lines the trap the question was built on. Numeric answers were computed by code from the question data.

Part A · Concept and interpretation (multiple choice)

Give the letter and a one-line reason, as in a written exam.

A1 1-1 CLO1-A

(b)

It reports what was observed in the people actually asked. Being from a sample doesn't make a statement inferential; **generalising** beyond the sample does.

A2 1-1 CLO1-A

(b)

The claim moves from the sample to all Muscat households, so it is a generalisation from sample to population: inferential.

A3 1-1 CLO1-A

(b)

Every member of the defined population was measured: a census. A measure of a whole population is a parameter, and reporting it is descriptive.

A4 1-2 CLO1-B

(b)

An ID is a number used only as a label: arithmetic on it is meaningless, so it is qualitative (nominal).

A5 1-2 CLO1-B

(c)

Differences between years are meaningful (2020 is 5 years after 2015), but year 0 does not mean "no time", so there is no true zero.

A6 1-2 CLO1-B

(b)

Time *duration* has a true zero, so ratios are valid. °C and IQ are interval; grades are ordinal.

A7 1-2 CLO1-B

(b)

The numbers only code an order. The gap from poor to fair is not known to equal the gap from good to excellent.

A8 1-2 CLO1-B

(c)

It is counted (discrete), and 0 patients means none (true zero), so it is ratio.

A9 1-3 CLO1-C

(b)

Every k-th member, k = 20, from a random start.

A10 1-3 CLO1-C

(b)

Some groups are chosen and all their members included: cluster. Stratified would take students from *every* college.

A11 1-3 CLO1-C

(b)

The respondents decide for themselves whether to take part, so those with strong opinions are over-represented.

A12 1-3 CLO1-C

(b)

A random sample differs from the population purely by chance: sampling error.

A13 1-3 CLO1-C

(b)

An instrument error is a nonsampling error. Taking a larger sample would not remove it.

A14 1-3 CLO1-C

(b)

Every stratum must be represented by a random sample. Sizes do not have to be equal (Dr. Amani's note).

Part B · Short exam problems by lesson

Classification with justification, Table D, systematic k, find-the-mistake.

B1 1-1 CLO1-A

(a) `all 3,500 first-year SQU students, N = 3500`. (b) `the 150 students asked, n = 150`. (c) `daily study hours: quantitative, continuous, ratio`. (d) `statistic`: it comes from the sample. (e) `inferential`: the conclusion is about all first-year students.

B2 1-1 CLO1-A

(a) `census`: every member of the population. (b) `survey`: a sample. (c) `census`: the population is defined as *her class*, and all of it is measured. Whether something is a census depends on how the population is defined.

B3 1-2 CLO1-B

(a) `qualitative, nominal` (b) `quantitative, discrete, ratio` (c) `quantitative, continuous, interval` (d) `qualitative, ordinal` (e) `quantitative, continuous, ratio`: a duration (f) `qualitative, nominal`: a label, even though it contains digits.

B4 1-2 CLO1-B

(a) `47.5–48.5 kg` (±0.5) (b) `3.65–3.75` (±0.05) (c) `0.1245–0.1255 s` (±0.0005)

Half of one unit of the *last recorded digit*, each way.

B5 1-2 CLO1-B · CLO3-C

Invalid. Temperature in °C is **interval**: 0 °C is not an absence of heat, so ratios are meaningless. Only the difference (+20 °C) is meaningful. On the Kelvin scale, a ratio scale, the temperatures are 293.15 K and 313.15 K, a ratio of only about 1.07.

B6 1-2 CLO1-B

Student B. Goals are **counted**, so the variable is discrete however large the counts get; 0 goals means none, so it is ratio. Continuous means *measured*, with infinitely many values between any two, which is not the same as "can be large".

B7 1-3 CLO1-C

N has 3 digits, so read 3-digit groups: 065 120 ~~783~~ 688 456 363 133.

Sample: **065, 120, 688, 456, 363, 133**. 783 is skipped because it is greater than 750.

B8 1-3 CLO1-C

(a) k = 250/12 = 20.83, rounded **down** to **20**. Sample: **7, 27, 47, 67, 87, 107, 127, 147, 167, 187, 207, 227**.

(b) 21, 42, ..., 231, then the 12th would be 252 > 250: only **11 patients**. Rounding up can leave the sample short, which is why the rule rounds down.

B9 1-3 CLO1-C · CLO3-C

An SRS might, by chance, contain too few of one gender to compare reliably. **Stratify by gender**: list males and females separately, then draw a random sample (for example by Table D or a computer) from **each** stratum. This guarantees both genders are represented; the two sample sizes may be equal or different.

B10 1-3 CLO1-C

(a) **convenience, biased**: early Saturday shoppers are not typical. (b) **volunteer, biased**: self-selected respondents with strong views. (c) **simple random (lottery method), unbiased**, provided the slips are well mixed.

B11 1-3 CLO1-C

P is **cluster**: cheaper and faster, with only 10 sites to visit. Q is **stratified** (schools as strata): every school is represented, but all 120 schools must be visited.

B12 1-3 CLO1-C

Tree 88 does not exist (N = 86). k should be 86/6 = 14.33, rounded **down** to 14. With start 13: **13, 27, 41, 55, 69, 83**.

B13 1-3 CLO1-C

The groups must have the same number of digits as N, which is **4**. 06512 = 6512 > 4998 would be rejected anyway. The correct start is **0651, 2078, 3688, 4563, …**.

B14 1-1 1-3 CLO1-A · CLO1-C · CLO3-C

(a) **inferential**: a general claim about a relationship. (b) **volunteer (self-selected)**. (c) The sample is biased (club members who chose to answer are not typical students), and 50 people is small. A survey can also only show an association: it doesn't prove that sleeping more *causes* a higher GPA.

Part C · Long structured questions

Midterm-style, multi-part. Marks in brackets.

C1 1-2 CLO1-B **[9 marks]**

(a) `qualitative, nominal` (b) `quantitative, continuous, ratio` (c) `quantitative, continuous, ratio` (d) `qualitative, ordinal` (e) `quantitative, interval` : a date has no true zero (f) `quantitative, discrete, ratio` (g) `qualitative, nominal` : a label (h) `quantitative, continuous, interval` (i) `qualitative, nominal`

C2 1-1 1-2 1-3 CLO1-A · B · C **[10 marks]**

1. Population `all licensed drivers in Oman` ; sample `the 1,200 drivers asked` ; variable `daily driving time: quantitative, continuous, ratio` .

2. Stratified: divide drivers by governorate (or by experience: under 5, 5–10, over 10 years) and take a random sample from **each** group. Cluster: treat, say, licensing centres or wilayats as clusters, randomly choose some, and survey **all** drivers registered there.

3. Drivers misremember or exaggerate their time; data-entry errors; a badly worded question.

4. (i) `descriptive` (ii) `inferential` : an estimate for all Omani drivers.

C3 1-3 CLO1-C **[10 marks]**

1. 3-digit groups: 065 120 ~~783~~ ~~688~~ 456 363 133 087 ~~951~~ 323 ~~764~~ ~~512~~ ~~557~~ ~~815~~ ~~986~~ ~~508~~ 455. Sample: `065, 120, 456, 363, 133, 087, 323, 455` .

2. k = 480/8 = `60` . The first 2-digit group is 06, which lies in 01–60, so the start is 6: `6, 66, 126, 186, 246, 306, 366, 426` .

3. `Stratified by gender` : a random sample from the male list and one from the female list.

C4 1-2 CLO1-B · CLO3-C **[8 marks]**

(i) `nominal` : only "different from". (ii) `ordinal` : "different" and "greater than"; a 10-unit difference has no fixed meaning. (iii) `interval` : different, greater, 10 units more; **not** twice. (iv) `ratio` : all four, so OMR 800 is twice OMR 400.

Part D · True / false with justification

The justification is the mark.

D1 1-2 CLO1-B · CLO1-C

`True` . Nominal and ordinal data are categories (qualitative); numerical data with meaningful differences are interval or ratio.

D2 1-2 CLO1-B · CLO1-C

`False` . The number of children is discrete and ratio: 0 means none, and 4 is twice 2.

D3 1-3 CLO1-B · CLO1-C

`True` . Sampling error comes from measuring only part of the population. A census can still have nonsampling errors such as recording mistakes.

D4 1-3 CLO1-B · CLO1-C

`False` . A defective instrument or a biased question gives wrong data at any sample size.

D5 1-3 CLO1-B · CLO1-C

`False` . Only the randomly chosen clusters do, with all their members. "Some from every group" describes stratified sampling.

D6 1-3 CLO1-B · CLO1-C

`False` . Dr. Amani's note: the sizes need not be equal, e.g. 12 from Statistics and 24 from Mathematics.

D7 1-2 CLO1-B · CLO1-C

`False` . They are labels; averaging them is meaningless. Qualitative, nominal.

D8 1-3 CLO1-B · CLO1-C

False. Size does not fix how subjects were chosen. A large biased sample is still biased.

Part E · Dr. Ahmed's review self-study exercises

The two exercises that close his deck.

E1 1-2 CLO1-B

(a) ratio (b) ordinal (c) nominal (d) interval

E2 1-2 CLO1-B

(a) qualitative, nominal (b) quantitative (continuous), ratio (c) qualitative, ordinal (d) quantitative, interval: dates have meaningful differences but no true zero (Dr. Ahmed's interval list).

Part F · In-Class Quiz I simulation, Chapter 1 part (15 marks · 15 minutes)

Closed book.

F1 1-1 CLO1-A

Generalising from samples to populations, performing estimations and hypothesis tests, determining relationships among variables, and making predictions. Example: the Ministry of Health surveys a sample of hospitals in November to predict next November's national flu cases.

F2 1-2 CLO1-B

(a) quantitative, discrete, ratio (b) qualitative, ordinal (c) quantitative, interval (d) quantitative, continuous, ratio

F3 1-3 CLO1-C

(a) systematic (b) cluster (c) stratified (d) simple random

F4 1-3 CLO1-C

k = 92/7 = 13.14, rounded down to 13. Sample: 9, 22, 35, 48, 61, 74, 87.

F5 1-2 CLO1-B

7.245–7.255 g and 119.5–120.5 cm

Chapter 1 study set: 7 files. One merged source built from Dr. Amani Al-Rubkhi's Fall 2026 notes (course coordinator) and Dr. Ahmed Al-Hadhrami's slide pack, aligned to the SQU STAT 2101 course outline (Fall 2026). Textbook: Bluman, *Elementary Statistics*, 10th ed.

00 · Exam brief · 01 · Lesson 1-1: Descriptive and Inferential Statistics · 02 · Lesson 1-2: Variables and Types of Data · 03 · Lesson 1-3: Data Collection and Sampling · 04 · Revision pack · 05 · Exam-style questions · **06 · Full worked answers**